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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B. Tech (CBCS) Degree Examinations
December 2025

IS2002-1 – MACHINE LEARNING FOUNDATIONS
(For ISE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is an application of machine learning?
A) Predicting stock prices
B) Building bridges
C) Writing software code
D) Designing hardware components
2. In reinforcement learning, what does an agent seek to maximize?
A) Computational speed
B) Data accuracy
C) Labelled datasets
D) Cumulative reward
3. Which of the following represents a challenge in machine learning?
A) Increasing computational power
B) Managing error and noise in data
C) Designing hardware
D) Writing software code
4. What is the learning curve in machine learning?
A) A measure of model complexity
B) A mathematical formula for learning rates
C) A graphical representation of model performance over time
D) A method for data transformation
5. Which distance metric calculates the straight-line distance between two points in Euclidean space?
A) Manhattan Distance
B) Euclidean Distance
C) Minkowski Distance
D) Cosine Similarity
6. What does a decision tree model do?
A) Divides data into branches to make predictions based on feature values
B) Groups similar data points into clusters
C) Reduces dimensionality of the data
D) Calculates the distance between data points
7. What is polynomial regression used for?
A) Reducing the number of features
B) Modeling non-linear relationships by including higher-degree terms
C) Classifying data into distinct categories
D) Clustering similar data points
8. In the K-Nearest Neighbors (KNN) algorithm, what does the parameter 'K' represent?
A) The distance metric used
B) The number of features in the dataset
C) The number of classes in the dataset
D) The number of nearest neighbors to consider
9. What is the goal of linear regression?
A) To find the line that best fits the data points
B) To classify data into categories
C) To cluster similar data points
D) To transform data into a different format
10. How does the k-Means algorithm initialize cluster centroids?
A) Randomly
B) Using the mean of all data points
C) Based on the median data point
D) By choosing the farthest data points

11. Which clustering technique requires a merging approach?
 A) Partitional B) Hierarchical
 C) Naive Bayes D) None of the mentioned
12. What is DBSCAN primarily used for in data mining?
 A) Regression analysis B) Clustering spatial data
 C) Dimensionality reduction D) Classification
13. How does DBSCAN handle noise points in a dataset?
 A) By assigning noise points to the nearest cluster B) By removing noise points from the dataset
 C) By ignoring noise points during clustering D) By treating noise points as a separate cluster
14. Which of the following is the first step in the Apriori algorithm?
 A) Generate candidate itemsets B) Calculate confidence
 C) Prune non-frequent itemsets D) Generate frequent itemsets
15. Which of the following is an advantage of the Naïve Bayes classifier?
 A) It works well with small datasets only B) It can handle continuous and categorical data well.
 C) It does not require independence of features. D) It is highly computationally expensive.
16. What is the main purpose of the activation function in a neural network?
 A) To scale the input data B) To introduce non-linearity into the model
 C) To adjust the learning rate D) To compute the loss
17. Random Forest can handle which of the following types of data?
 A) Only numerical data B) Only categorical data
 C) Both numerical and categorical data D) Only text data
18. Which of the following algorithms is a popular example of boosting?
 A) K-Nearest Neighbors B) AdaBoost
 C) Support Vector Machine D) Linear Regression
19. For a given dataset, increasing the number of bins in a histogram estimator typically results in
 A) A smoother density estimate B) A more jagged density estimate
 C) No change in the density estimate D) A less accurate estimate of the density
20. Which of the following metrics is affected the most by class imbalance?
 A) Accuracy B) Recall
 C) Precision D) F1 Score

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Explain the following with suitable examples
 I. Supervised Learning,
 II. Un supervised learning
 III. Reinforcement Learning
 b) A dataset is given:

Study Hours (X)	Exam Score (Y)
1	40
2	50
3	65
4	70
5	85

- (a) Fit a Simple Linear Regression model (find slope and intercept).
 (b) Predict the exam score for a student who studies 6 hours.

8 L3 1 2

2. a) Compute the following similarity/dissimilarity measures:
 (a) Euclidean distance between points (2,3) and (5,7).
 (b) Cosine similarity between vectors $A = (1, 1, 0, 1)$ and $B = (0, 1, 1, 1)$.
 (c) Jaccard similarity for binary vectors $A = (1, 0, 1, 1, 0)$ and $B = (1, 1, 1, 0, 0)$.
 b) Discuss the bias-variance tradeoff with the help of a neat diagrams of the learning curve.

8	L3	2	2
8	L2	1	1

3. a) Consider the dataset:

Outlook	Temperature	Humidity	Play
Sunny	Hot	High	No
Sunny	Hot	Normal	No
Overcast	Mild	High	Yes
Rain	Mild	Normal	Yes
Rain	Cool	Normal	Yes

- (a) Calculate the entropy of the dataset.
 (b) Compute the information gain for attributes Outlook and Humidity
 (c) Select the best attribute for the root node.
 b) Write a program to implement k-Nearest Neighbor classification algorithm on the iris flower dataset and visualize the results.

8	L3	2	2
8	L2	2	1

Unit – II

4. a) Explain the working of a Naïve Bayes classifier with a suitable numerical example.
 b) With the help of diagrams, explain the difference between linear SVM and non-linear SVM using kernel trick.

8	L3	3	1
8	L2	3	2

5. a) A market basket dataset has the following transactions:

T1: {Milk, Bread, Butter}
 T2: {Bread, Diaper, Beer, Eggs}
 T3: {Milk, Diaper, Bread, Butter}
 T4: {Bread, Milk, Diaper, Beer}
 T5: {Milk, Bread, Butter, Diaper}

- (a) Generate frequent itemsets using Apriori Algorithm with min support = 0.6.

- (b) Derive at least two association rules with min confidence = 0.7

8	L3	3	3
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- b) Compare and contrast the following clustering approaches:

- (a) K-means vs Hierarchical clustering
 (b) DBSCAN vs K-means

8	L2	3	1
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6. a) Perform Agglomerative Hierarchical Clustering on the following 1D data points: {2, 4, 5, 10, 12} using single linkage (minimum distance).

- (a) Show the sequence of merges.
 (b) Draw the dendrogram

8	L3	4	2
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- b) Explain the working of the FP-Growth Algorithm. Compare its efficiency with Apriori algorithm in terms of candidate generation, memory usage, and time complexity.

8	L2	4	1
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Unit – III

7. a) Explain the architecture of a perceptron. Discuss how it is trained for classification.
 b) Describe the Backpropagation algorithm with a neat diagram. Why is it widely used in neural networks?

8	L2	5	1
8	L2	5	1

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8. a) Explain the working of the Random Forest algorithm. How does it overcome the overfitting problem of decision trees?
- b) Explain Precision, Recall, F1 Score, and AUC curve with suitable examples

8	L2	5	1
8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
